CS182: Introduction to Machine Learning
Lecture 05: Bayesian Decision Theory

Lecture Quiz and Answer Copy

Scope: Bayesian_Decision_Theory.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Name: _____ **Student ID:** _____

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Answer sheet — Section A

1	2	3	4	5	6

7	8	9	10	11	12

Answer sheet — Section B

13	14	15	16	17	18

19	20	21	22	23	24

Section A. True/False 12 marks

1. A class-conditional density $p(x \mid C_i)$ must lie between 0 and 1 at every x.

2. With unequal misclassification costs, a class whose posterior is below one half can still be the minimum-risk classification action.

3. Every point satisfying $g_i(x) = g_j(x)$ belongs to a decision boundary between C_i and C_j.

4. Replacing every discriminant by $a(x)g_i(x) + b(x)$, using the same $a(x) > 0$ and $b(x)$ for all classes, preserves the entire set of maximizing classes.

5. Under 0-1 loss, if no features are observed, always predicting a class with the largest prior probability minimizes the probability of error.

6. Applying a different strictly increasing function to each class discriminant always preserves the classifier.

7. Because the evidence $p(x)$ cancels when comparing classes, it can also be omitted when reporting normalized posterior probabilities.

8. Under 0-1 loss, choosing the largest class likelihood always gives the Bayes decision, even with unequal class priors.

9. Randomly mixing classification actions at a fixed x can give conditional risk strictly below the smallest risk of those actions.

10. If two classes tie for the largest posterior probability, choosing either one has the same conditional 0-1 risk.

11. With unit error cost and rejection cost λ, classification is strictly preferred to rejection exactly when the largest posterior exceeds $1 - \lambda$.

12. Naive Bayes requires the features to be independent in the unconditional feature distribution.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

13. For $x \in \mathbb{R}$, define $g_1(x) = 1 - x^2$, $g_2(x) = 0$, and $g_3(x) = 1/2$. Use the largest score and allow ties on boundaries. Select every correct statement.

(A) Class C_2 has a nonempty decision region.

(B) The region where C_3 is the unique winner has two disconnected components.

(C) The actual class changes occur at $x = \pm 1/\sqrt{2}$.

(D) The equation $g_1 = g_2$ produces $x = \pm 1$, but neither point is a decision boundary.

14. Let the true joint distribution be known, and use 0-1 loss. Select every correct statement about using features.

(A) The Bayes error using X is no larger than the error of always predicting a class with largest prior.

(B) Any rule that depends on X must outperform every constant rule.

(C) If each observed input identifies its class with posterior probability 1 almost surely, the Bayes error is zero.

(D) If the posterior equals the prior almost surely, observing X cannot improve the minimum error.

15. A three-class posterior is $(0.46, 0.34, 0.20)$. Correct classifications cost 0 and mistakes cost 1. Select every correct statement about a constant rejection cost.

(A) At rejection cost 0.55, acceptance requires the largest posterior to exceed 0.55.

(B) At rejection cost 0.50, rejection is strictly preferable to every class action.

(C) At rejection cost 0.55, predicting C_1 is strictly preferable to rejection.

(D) Increasing the rejection cost cannot enlarge the set where rejection is strictly optimal.

16. Assume all discriminants are finite. Which transformations preserve the full maximizing-class set at every x, for arbitrary original scores?

(A) Multiply every score by the same negative constant.

(B) Add a different constant b_i to each class score.

(C) Apply one common strictly increasing function f to every score.

(D) Replace every $g_i(x)$ by $a(x)g_i(x) + b(x)$, with common $a(x) > 0$ and $b(x)$.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

17. Consider any finite loss matrix λ_{ik}. Which modifications preserve all minimum-risk actions for every posterior?

(A) Add one state-dependent number b_k to every entry in column k.

(B) Replace risks by their negatives and choose the largest resulting score.

(C) Multiply the entire loss matrix by one positive constant.

(D) Add a different arbitrary constant to every row.

18. For two classes with positive priors and positive likelihoods, let $P(C_1) = 1/4$, $P(C_2) = 3/4$, and $L(x) = p(x \mid C_1)/p(x \mid C_2)$. Under 0-1 loss, select every correct statement.

(A) For the given priors, a log discriminant is $\log L(x) - \log 3$.

(B) Predicting C_1 is strictly optimal when $L(x) > 3$.

(C) The evidence is needed to determine which posterior is larger.

(D) Replacing the priors by 0.1 and 0.9 changes the likelihood-ratio threshold to 9.

19. Which statements about the naive Bayes assumption are correct?

(A) It is sufficient that the features be mutually independent conditional on each class.

(B) Unconditional independence alone guarantees the required class-conditional factorization.

(C) The class prior is unnecessary once the likelihood is factorized.

(D) The factorization makes $P(C_i \mid x)$ equal to $\prod_j P(C_i \mid x_j)$ without any normalization or prior correction.

20. For three classes, let the priors be $(0.2, 0.3, 0.5)$ and the class likelihoods at one observed x be $(0.9, 0.4, 0.2)$. Under 0-1 loss, select every correct statement.

(A) The Bayes action predicts C_1.

(B) The minimum conditional probability of error is 0.45.

(C) The posterior probability of C_1 is 0.60.

(D) The evidence is $p(x) = 0.40$.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

21. A posterior is $(0.4, 0.4, 0.2)$, unit error loss is used, and rejection costs 0.6. Select every correct statement.

(A) Any random mixture of rejection and predictions of C_1 or C_2 also has risk 0.6.

(B) Predicting C_3 is also optimal because all posterior values are below 0.6.

(C) The minimum conditional risk is 0.6.

(D) Rejection is the unique risk-minimizing action.

22. Write $q = P(C_1 \mid x) = 0.55$. Correct decisions cost 0; predicting C_1 when C_2 is true costs 8, and predicting C_2 when C_1 is true costs 2. Select every correct statement.

(A) Predicting C_1 has risk 3.6.

(B) Predicting C_2 minimizes conditional risk.

(C) Without rejection, predicting C_1 is strictly optimal when $q > 0.8$.

(D) The larger posterior must determine the minimum-risk action.

23. At a fixed input, suppose the risks of three available actions are $0.2, 0.5, 0.2$. Select every correct statement about randomized actions.

(A) Assigning positive probability to the second action can still yield risk below 0.2.

(B) Every minimum-risk randomization must put zero probability on the second action.

(C) A half-and-half mixture of the first and third actions has risk 0.2.

(D) The existence of two minimizing actions forces a unique optimal randomization.

24. At an observed point, two continuous class densities have values 4 and 1, and their priors are 0.1 and 0.9. Under 0-1 loss, select every correct statement.

(A) The evidence density is 1.3.

(B) The posterior of C_1 is $4/13$.

(C) The Bayes action predicts C_2.

(D) The larger likelihood proves that C_1 is the Bayes choice.

Section C. Proofs and Formula Derivations

25. Why pointwise decisions minimize total risk **[10 marks]**

(a) Let A be a finite action set, let λ_{ak} be finite nonnegative losses, and let X have density or mass function $p(x)$. Define a deterministic rule $\delta(x)$. Starting from $E[\lambda_{\delta(X),C}]$, derive the total-risk formula and prove that choosing a minimum-conditional-risk action at each x minimizes total risk. State the almost-sure qualification. **[4 marks]**

(b) Specialize to K classification actions and 0-1 loss. Derive the MAP rule and the Bayes error expression in terms of the largest posterior. **[3 marks]**

(c) Prove that this Bayes error is at most $1 - \max_i P(C_i)$, the best constant-rule error. Explain why this does not guarantee that every fitted classifier improves on the constant rule. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

26. Asymmetric errors and rejection thresholds **[10 marks]**

(a) For two classes, write $q = P(C_1 \mid x)$. Correct decisions cost 0, predicting C_1 when C_2 is true costs $a > 0$, and predicting C_2 when C_1 is true costs $b > 0$. Derive the optimal posterior threshold without rejection and the equivalent likelihood-ratio threshold when priors and likelihoods are positive. **[4 marks]**

(b) Add a reject action with constant loss $r > 0$ for either true class. Derive the interval where rejection is strictly optimal and determine when that interval has positive width. **[3 marks]**

(c) Return to K classes, unit cost for any error, and reject cost λ. Derive the acceptance rule. Then show that if $\lambda > 1 - 1/K$, rejection cannot be strictly optimal at any input. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

27. Equivalent discriminants and the role of strict monotonicity **[10 marks]**

(a) Prove that applying one common strictly increasing function to all finite discriminants preserves their full argmax set. Explain why merely nondecreasing is insufficient for preserving the argmax set, using a concrete counterexample. **[4 marks]**

(b) For $p(x) > 0$ and positive class joint scores, derive the equivalence of posterior, prior-weighted likelihood, and log prior-weighted likelihood discriminants. Identify the qualification needed for zero joint scores. **[3 marks]**

(c) For two classes with positive priors and likelihoods, derive a single scalar log discriminant. Show how its sign determines the 0-1 decision, and explain why a negative-risk discriminant rather than posterior probability is needed for arbitrary losses. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

28. What a decision boundary actually represents **[10 marks]**

(a) Suppose $g_i(x) = w_i^\top x + b_i$ on $\mathbb{R}^d$. Show that the weak decision region $\mathcal{R}_i = \{x : g_i(x) \geq g_j(x) \text{ for all } j\}$ is convex. Describe the supporting equality between two competing classes. **[4 marks]**

(b) Give an explicit two-class example on $\mathbb{R}$ with a disconnected decision region, and derive the regions and boundaries. Use $g_1(x) = x^2 - 4$ and $g_2(x) = 0$. **[3 marks]**

(c) For $g_1(x) = 2 - x^2$, $g_2(x) = 0$, and $g_3(x) = 1$, identify all pairwise equality points and determine which are actual decision boundaries. Use the result to disprove the claim that every discriminant equality is a boundary. **[3 marks]**

Student response

(a)

(b)

(c)

Section D. Applications

29. A sensor with asymmetric mistakes and a review action **[12 marks]**

(a) A sensor reports A, B, or C. The class priors are $P(C_1) = 0.35$, $P(C_2) = 0.65$, and the class-conditional probabilities for (A, B, C) are $(0.50, 0.30, 0.20)$ in C_1 and $(0.10, 0.40, 0.50)$ in C_2. Compute each report probability and each posterior $q_s = P(C_1 \mid s)$. **[4 marks]**

(b) Correct classifications cost 0. Predicting C_1 when C_2 is true costs 3; predicting C_2 when C_1 is true costs 9. A human review action costs 1.7 regardless of class. Compute all three conditional risks for each report and choose the optimal action. **[4 marks]**

(c) Compute the expected loss of your review-enabled rule. Compare it with the best rule forced to classify every report and with the MAP rule that ignores the asymmetric costs. **[4 marks]**

Student response

(a)

(b)

(c)

Section D. Applications

30. Conditional independence, posterior decisions, and rejection **[12 marks]**

(a) Two binary features are independent conditional on class. Let $P(C_1) = 0.4$, $P(C_2) = 0.6$; within C_1, the probabilities $P(X_1 = 1 \mid C_1)$, $P(X_2 = 1 \mid C_1)$ are $0.8, 0.3$; within C_2, they are $0.25, 0.7$. For every binary pair $(00, 01, 10, 11)$, compute the two joint class-feature scores, give the MAP class, and find the overall 0-1 Bayes error. **[4 marks]**

(b) Add a rejection action costing 0.30, with unit misclassification costs. Determine which feature pairs are accepted and compute the total expected risk including rejected inputs. **[4 marks]**

(c) Are X_1 and X_2 independent without conditioning on class? Establish your answer by computing the necessary marginal and joint probabilities. **[4 marks]**

Student response

(a)

(b)

(c)